

LRSC Structural Mechanism Audit v1.1

Prince Upadhyay - Independent Research

Audited candidate | Locked benchmark M=99, k=49, A=0.50, theta=0.08

Executive result

The locked operator reproduces 81 active sites and full 50-channel closure at relative L2 error 2.381e-16. The exact modular-folding identity and exact Gram reduction are established algebraically. The channel family has numerical rank 41 with a large singular-value cliff. A single cardinality ≤ 10 scaled MILP returns the known K10 edge-block optimum with reported MIP gap 0.0; direct reconstruction gives 0.001039820619, while the explicit K11 witness gives 0.000962369128.

1. Exact modular folding

For odd $M=2k+1$, $2k$ is congruent to -1 modulo M . Therefore $(2s)k$ is congruent to $-s$ and $(2s+1)k$ is congruent to $k-s$. After conjugate balancing, the orbit is:

$$k, 1, k-1, 2, k-2, 3, k-3, \dots$$

For $M=99$ and $k=49$: 49, 1, 48, 2, 47, 3, 46, 4, Channel 3 is therefore the balanced representative of the sixth modular harmonic because $6 \cdot 49 = 294$ is congruent to -3 modulo 99.

2. Gram reduction

Let $H_{rs} = \text{Re}\langle V_r, V_s \rangle$ and let z be the binary channel-selection vector. Since the exact target d is the sum of all channel vectors, the residual identity is:

$$E^2(z) = (1-z)^T H (1-z) / (1^T H 1)$$

This is an exact algebraic reduction of the fixed-unit spectral-compression problem to one 50 x 50 Gram matrix.

3. Rank mechanism

The locked mask has 81 active sites and reflection symmetry, giving 41 physical reflection orbits. This proves a structural rank upper bound of 41 for the pre-stencil reflection-symmetric family. Fresh construction gives numerical rank 41, with $\sigma_{41}=2.290338e-04$ and $\sigma_{42}=3.625328e-16$. The 41 physical representatives also have numerical rank 41.

The exact lower-bound route is a Chebyshev/Vandermonde argument: $C_{ri} = \cos(2\pi r \cdot i / 99) = T_r(x_i)$, with 41 distinct x_i . The cosine evaluation matrix is nonsingular. For the actual channel family, a fully exact theorem additionally requires rigorous nonvanishing certification of the channel scale factors; the present numerical minimum $|\hat{j}[r]|$ for $r=0..40$ is 2.028046e-04.

4. Cardinality ≤ 10 computation

Constraint	Selected channels	Direct relative L2	Status
$K \leq 10$	(0, 1, 42, 43, 44, 45, 46, 47, 48, 49)	0.001039820619	FAILS 0.001
$K = 11$ witness	(0, 1, 3, 42, 43, 44, 45, 46, 47, 48, 49)	0.000962369128	PASSES 0.001

The ≤ 10 optimization uses binary product linearization, objective scaling by $1e12$, cardinality $\sum(z) \leq 10$, and HiGHS through SciPy MILP with requested relative MIP gap 0.0. The returned solution reported gap 0.0. The final error is recomputed directly from the original complex channel vectors. This is strong computational global evidence, but it is not labeled an exact-rational proof.

5. Channel-3 interpretation

Two mechanisms must remain separate. Modular arithmetic explains why channel 3 belongs to the carrier's folded harmonic family. Gram geometry explains why it is selected after the K10 state: it gives the largest

marginal reduction of the remaining residual among excluded channels. No numerological interpretation is used.

6. Claim boundary

This audit concerns the locked LRSC mathematical operator only. It does not establish a universal physical dimension, a law of nature, or any 3-6-9 interpretation. LRSC v1.0 remains untouched. v1.1 should be frozen only at the evidence level stated here unless the remaining exact nonvanishing and solver-independent lower-bound certificates are added.